MIL-L-11647D
31 August 1972
SUPERSEDING
MIL-L-11674C
17 December 1963

MILITARY SPECIFICATION

LEVER, BALK DEPRESSING:

CLAW-BAR TYPE.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a claw-bar type, balk depressing lever to be used in the erection of fixed and floating bridges having deck-balk superstructures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601 - Boxes, Wood, Cleated-Plywood.

Military

MIL-T-704 - Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595 - Colors.

FSC 5420

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

DRAWINGS

ME

D13219E3956	- Lever, Balk Depressing: Claw-Bar Type.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure
D2.0 - Welded Highway and Railway Bridges.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The claw-bar type, balk depressing lever consisting of a modified crowbar, hooks, chains, and attachments shall be as shown on Drawing D13219E3956 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one balk depressing lever for examination and test within the time frame specified (see 6.2), to prove, prior to starting production, that his production methods will produce balk depressing levers that comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required.

3.5 Performance. The balk depressing lever shall withstand a load of 450 pounds applied 6 inches from the end of the handle while the lever is engaged at the point without permanent deformation in the handle or point, or failure in the welds.

3.6 Identification marking. The balk depressing lever shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

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3.7 Treatment and painting. The balk depressing lever shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34087 in accordance with FED. STD. No. 595.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel used in the fabrication of the balk depressing lever shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

3.8.2 Welders and welding.

3.8.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welder Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D2.0 - Welded Highway and Railway Bridges". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. The balk depressing levers which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and test in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction balk depressing lever shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The balk depressing lever shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The balk depressing lever shall be examined as specified herein for the following defects:

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Major

101. Nonconformance to the dimensions shown on the drawing.
102. Material not as specified.
103. Workmanship not as specified.
104. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Cleaning, treatment, and painting not as specified.

4.5.2 Performance test. Apply a load of 450 pounds 6 inches from the end of the handle while the lever is engaged at the point as shown in figure 1. Apply the load for not less than 2 minutes. Any permanent deformation in the handle or point or any failure in the welds shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Containers not as specified for level A or B.
106. Chains not secured to the bar as specified for level A or B.
107. Blocking, bracing, and anchoring not applied as specified for level A or B.
108. Strapping not zinc-coated for level A.
109. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The chains shall be secured to the bars with wire ties. The levers shall be packed together in boxes conforming to PPP-B-601, overseas type, style optional, grade B. Blocking, bracing, and anchoring shall

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be applied in accordance with MIL-STD-1186. Boxes exceeding 200 pounds shall be modified with skids in accordance with the box specification. Strapping shall be zinc-coated and on modified boxes shall be placed parallel with and adjacent to the skids.

5.1.2 Level B. The levers shall be packed as specified in 5.1.1, except boxes shall be domestic type and strapping need not be zinc coated.

5.1.3 Level C. The levers shall be packed or prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lever is intended for use in depressing the balk to facilitate placement of the pins during the erection of fixed and floating bridges having deck balk superstructure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production balk depressing levers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish balk depressing levers conforming to this specification.

Custodians:

Army - ME

Navy - MC

Preparing activity:

Army - ME

User activity:

Project No. 5420-0092

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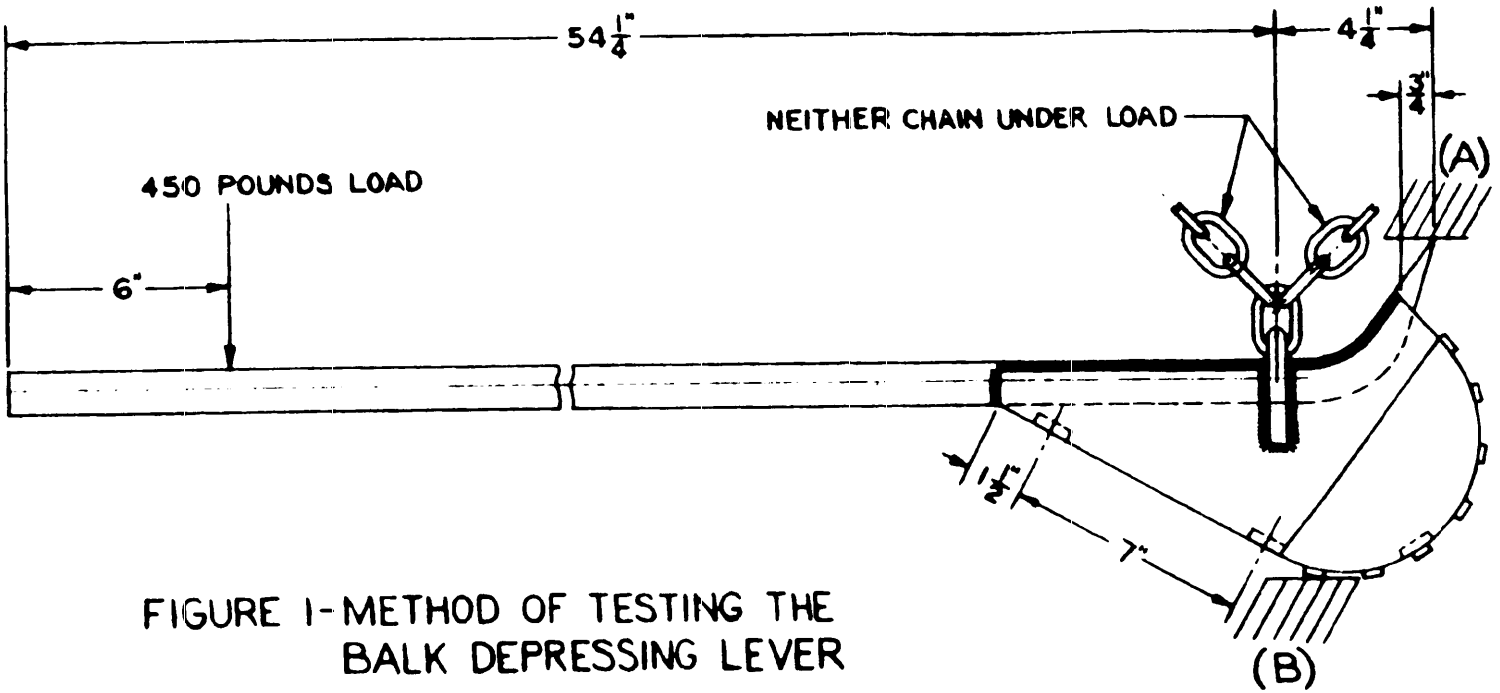


FIGURE 1-METHOD OF TESTING THE BALK DEPRESSING LEVER

NOTE:
THE POINT AND LUG WILL BE UNDER REACTIVE LOADS AT SUPPORTING SURFACES(A) AND (B), RESPECTIVELY.

BX-2131

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-L-11647D, Lever, Balk Depressing: Claw-Bar Type

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZD
FORT BELVOIR, VIRGINIA 22060

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